CS582 Project Proposal

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Motivation

Non-coding regions of DNA, though they do not encode proteins, constitute approximately 98.5% of the human genome and play a crucial role in regulating gene expression (Boland, 2017). These regions govern when, where, and how genes are expressed, and disruptions within them can lead to a range of genetic disorders and diseases. Despite their importance, non-coding regions remain comparatively understudied and difficult to interpret, especially when mutations alter their regulatory functions.

Within the non-coding genome, promoters, enhancers, and untranslated regions (UTRs) are of particular interest. Promoters control transcription initiation, enhancers activate gene expression over long genomic distances, and UTRs influence mRNA stability and translation efficiency. Understanding how variants within these elements affect gene regulation is essential for improving our ability to predict disease risk and uncover new therapeutic targets.

By focusing on regulatory variants in human tissues, this research aims to deepen our understanding of the genetic mechanisms underlying disease and inform future efforts in genetic disorder prevention.

Problem definition: data, method, challenges, etc.

Overall, we plan to test which models are best at predicting the regulatory impact of non-coding variants across different genomic contexts, such as enhancers, promoters, and untranslated regions. We are also interested in assessing how well modern deep learning models, such as transformers, perform on non-coding regulatory variants over traditional annotation-based models.

We will be using three different models to assess their performance in creating the predictions. These include CADD (Combined Annotation Dependent Depletion) (Rentzsch et al., 2019), which is a Classical ensemble model, DeepSEA, which is a Deep learning CNN (Zhou & Troyanskaya, 2015), and Enformer, which is a transformer-based deep learning model (Avsec et al., 2021).

We will be exploring labeled variant data for non-coding regions such as promoters, enhancers, and UTRs. Variant Call Format is a text file that contains the chromosome, position, reference allele, and alternate allele for each variant. CADD will only need the annotated regulatory variants, but DeepSEA and Enformer will need the human reference genome since we will be focusing on human tissue. We will be using sources such as ClinVar for labeled variants and filtering for non-coding variants (promoters, enhancers, and UTRs) with annotation tools, and then we will use the FASTA file for the human reference genome hg38.

Here are some potential challenges that we might face:

- Data quality: Mismatches in non-coding data, such as not matching the reference or genome inconsistencies, can disrupt computation and bias performance.
  - Solution: Check each input sequence against the reference, and fix or drop discordant sites while keeping true biological variants unchanged.
- Scale of computation: Non-coding regions make up ~ 98% of the genome, creating extremely large computational costs for the models.
  - Solution: Prioritize non-coding regions such as promoters, enhancers, and UTRs rather than the complete non-coding regions.

Planned Timeline

Task	Description	Date	Published	Assigned
Review Literature	Survey 3-5 papers on non-coding variant prediction	Oct 31	Done	Ellen Yang Kavya Puranam
Submit Preproposal	1-2 pages proposal	Oct 31	Done	Ellen Yang Kavya Puranam
Data Exploration and Cleaning	Curated non-coding genomic data with available datasets	Nov 3	In Progress	Ellen Yang Kavya Puranam
Preliminary Data Statistics	Assemble available non-coding genomic data	Nov 5	Not Started	Ellen Yang Kavya Puranam
Preliminary Data Visualizations	Create plots and images	Nov 7	Not Started	Ellen Yang Kavya Puranam
1st Model Architecture Setup	Implement CADD	Nov 12	Not Started	Ellen Yang
1st Model Run	Run model with datasets and benchmark performance	Nov 14	Not Started	Ellen Yang
2nd Model Architecture Setup	Implement DeepSEA	Nov 19	Not Started	Kavya Puranam
2nd Model Run	Run model with datasets and benchmark performance	Nov 21	Not Started	Kavya Puranam
3rd Model Architecture Setup	Implement Enformer	Nov 26	Not Started	Ellen Yang

Planned Timeline				
Task	Description	Date	Published	Assigned
3rd Model Run	Run model with datasets and benchmark performance	Nov 28	Not Started	Ellen Yang
In-Class Presentation	Prepare 5-10 minutes presentation	Dec 12	Not Started	Ellen Yang Kavya Puranam
Submit Final Report	8-10 pages report excluding reference, NeurIPS overleaf template	Dec 16	Not Started	Ellen Yang Kavya Puranam

Evaluation and goals

We plan to benchmark three models, CADD, DeepSEA, and Enformer, under metrics AUROC, AUPRC, Spearman's rank correlation, Accuracy, F1, Precision, and Recall (Steyerberg et al., 2010). These benchmarks aim to evaluate how non-coding regions relate to mutation effects and to identify the best-performing model. Comparing performance across measures will answer our research questions and yield recommendations for non-coding variant prediction.

While many deep learning models evaluate mutation effects in coding regions, there lack of rigorous work on how variants in promoters, enhancers, and other non-coding elements can alter gene expression. Our project aims to fill this gap by focusing on the non-coding region and prioritizing regulatory variants that may resolve cases left unsolved by coding-only expression. Ultimately, we seek to show how non-coding region mutation can guide diagnostics of disease.

Related work:

[1] Richard Boland C. (2017). Non-coding RNA: It's Not Junk. Digestive diseases and sciences, 62(5), 1107–1109. https://doi.org/10.1007/s10620-017-4506-1
[2] Rentzsch, P., Witten, D., Cooper, G. M., Shendure, J., & Kircher, M. (2019). CADD: predicting the deleteriousness of variants throughout the human genome. Nucleic acids research, 47(D1), D886–D894. https://doi.org/10.1093/nar/gky1016
[3] Zhou, J., & Troyanskaya, O. G. (2015). Predicting effects of noncoding variants with deep learning–based sequence model. Nature Methods, 12(10), 931–934. https://doi.org/10.1038/nmeth.3547
[4] Avsec, Ž., Agarwal, V., Visentin, D., Ledsam, J. R., Grabska-Barwinska, A., Taylor, K. R., Assael, Y., Jumper, J., Kohli, P., & Kelley, D. R. (2021). Effective gene expression prediction from sequence by integrating long-range interactions. Nature Methods, 18(10), 1196–1203. https://doi.org/10.1038/s41592-021-01252-x

[5] Steyerberg, E. W., Vickers, A. J., Cook, N. R., Gerds, T., Gonen, M., Obuchowski, N., Pencina, M. J., & Kattan, M. W. (2010). Assessing the performance of prediction models: a framework for traditional and novel measures. Epidemiology (Cambridge, Mass.), 21(1), 128–138. https://doi.org/10.1097/EDE.0b013e3181c30fb2